Review Sheet 2: Managing Variability[1]

Suggested Answers

1. **A Petroleum Refinery***

An integrated petroleum company is considering expansion of its one unloading facility at its Australian refinery. Due to random variations in weather, loading delays, and other factors, ships arriving at the refinery to unload crude oil arrive according to the Poisson distribution (CV_a=1) with an average arrival rate of 8 ships per week. Service time is exponential with an average service rate of 12 ships per week (CV_S=1)

Mean arrival rate of ships to facility:

$\lambda = 8 \text{ ships / week}$

Mean service time per ship at facility (capacity of facility):

$\mu = \text{service rate} = 12 \text{ ships / week}$

Mean utilization rate of facility:

$\rho = 8 \text{ ships per week/ } 12 \text{ ships per week} = 0.666 \text{ (66.6\% busy)}$

Because arrivals are Poisson, $CV_A^2 = 1$, *and because service times are exponential,* $CV_S^2 = 1$.

Also note that $\tau = 1/\mu = 0.0833$ weeks.

2. Current performance measures:

   i. ***What is the average number of ships waiting to gain access to the single unloading facility?***

Little's Law tells us that

$L_q = \lambda W_q$,

so once we find W_q, *we will have* L_q.

$$W_q = \tau \frac{\rho}{(1-\rho)} \frac{1}{2}\left(CV_A^2 + CV_S^2\right)$$

$$= 0.0833 \textit{ (weeks)} \frac{0.666}{(1-0.666)} \frac{1}{2}(1+1)$$

$$= 0.167 \textit{ (weeks)}$$

and thus,

$$L_q = \lambda \times Wq = 8 \text{ ships/week} \times 0.167 \text{ weeks} = 1.336 \text{ ships}$$

[1] Problems are rated for difficulty: *= Relatively easy, for first review of concepts; **= Challenging, assumes good understanding of the concepts; ***=Advanced, assumes full understanding of the concepts and creativity in application.

ii. What is the average time a ship must wait before beginning to deliver its cargo to the refinery?

$W_q = 0.167(\text{weeks})$

iii. What is the average total time (waiting plus actual delivery) that a ship spends at the refinery?

$W = W_q + \tau = 0.167 + 0.0833 \approx 0.25 \text{ weeks}$.

b) The company has under consideration a second unloading berth that could be rented for $6,000 per week. The service time for this berth would also be exponential with the same service rate of 12 ships per week as the company's own berth. The opportunity cost of a ship not being at sea is $30,000 per week.

Since there are two servers, sharing the same load, the overall utilization of the facility will be decreased. We denote the number of servers by s.

Mean utilization rate of facility:

$$\rho = \frac{\lambda\tau}{s} = \frac{8 \text{ ships/week} \times 0.0833 \text{ weeks/ship}}{2} = 0.33 \quad \textit{(33\% busy)}$$

i. If the second berth is rented, what will be the average number of ships waiting?

$$W_q = \frac{\tau}{s}\frac{1}{1-\rho}\frac{1}{2}\left(CV_a^2 + CV_s^2\right)\rho^{(\sqrt{2(s+1)}-1)}$$

$W_q = 0.01$

$L_q = \lambda W_q = 8*0.01 = 0.08$

ii. What would be the average total time a ship would wait?

$W = \tau + W_q$

$W = 0.0833 + 0.01 = 0.0933 \; weeks$

iii. Is the benefit of reduced waiting time (in dollars) worth the rental cost for the second berth?

$$(0.25\,weeks - 0.0933\,weeks) \times \frac{\$30{,}000}{ship \times week} \times \frac{8\,ships}{week} \approx \$37{,}600/week \text{ vs } \$6{,}000/\text{week}$$

Yes, it is worth the investment given the reduction in the weekly cost of delay to ships, for an estimated gain per week of $31,600 (Note that, if we use fractions all the way, we will have 37,453 $/week instead of 37,600).

c) An alternative way to improve unloading facilities is for the company to modify the current operations to speed up the unloading process. Specifically, the new service time would still be exponential, but with a mean service rate of 15 ships per week. The incremental cost of the new operations would be $ 4,000 per week.

Mean arrival rate of ships to facility:

$\lambda = 8$ *ships per week*

Mean service time per ship at facility:

Only the mean service rate is given:

μ = *service rate* = *15 ships per week*

Note that the mean service rate is just the inverse of the mean service time:

$\mu = 1/\tau \;\Rightarrow\; \tau = 1/\mu = 0.0667$ *weeks per ship*

Mean utilization rate of facility:

$\rho =$ *8 ships per week / 15 ships per week = 0.53 (53% busy)*

i. Compute the (new) expected waiting time and total time for ships.

$$W_q = \tau \frac{\rho}{(1-\rho)} \frac{1}{2}\left(CV_A^2 + CV_S^2\right)$$

$$= 0.0667 \frac{.53}{(1-.53)} \frac{1}{2}(1+1)$$

$$= 0.075 \textbf{ weeks}$$

$$W = W_q + \tau = 0.075 + 0.067 = 0.14 \textbf{ weeks}.$$

ii. Is the benefit worth the incremental cost?

$$(0.25 \text{ weeks} - 0.14 \text{ weeks}) \times \frac{\$30{,}000}{\text{ship week}} \times \frac{8 \text{ ships}}{\text{week}} = \$26{,}400/\text{week benefit vs. } \$4{,}000/\text{week cost}$$

Yes, the benefit in reduced total time at the facility is worth the investment, for a weekly net gain of $22,400 (Note that we have 25,714 $/week instead of 26,400 when using fractions).

Please note that the rental of a second unloading berth (alternative in question (b) above) is preferable to modifying current operations (alternative in question (c) here).

iii. How would you answer i - ii above if, instead of speeding up the service rate of the facility, the new operations made the service time constant, but still with the old service rate of 12 ships per week? Assume that the incremental cost of this change would be $ 4,000 per week.

If the arrival rate is 8 ships per week and the mean service rate is 12 ships per week, then the utilization of the capacity is

ρ *= 8 ships per week/ 12 ships per week = 0.666 (66.6% busy)*

We use the same formula as in (ii), but now $CV_S^2 = 0$ *- Why? Since the service time is constant, there is no variability, and the coefficient of variation for service is zero. Then, we get,*

$$W_q = \tau \frac{\rho}{(1-\rho)} \frac{1}{2}\left(CV_A^2 + CV_S^2\right)$$

$$= 0.0833 \frac{0.666}{(1-0.666)} \frac{1}{2}(1+0)$$

$$= 0.0833 \text{ weeks}$$

Therefore $W = W_q + \tau = 0.0833 + 0.0833 = 0.167$ weeks.

Finally, we determine the change in the average total waiting time, to determine the economic benefit of the investment:

$$(0.25\,\textbf{weeks} - 0.167\,\textbf{weeks}) \times \frac{\$30{,}000}{\textbf{ship week}} \times \frac{8\,\textbf{ships}}{\textbf{week}} = \$20{,}160/\textbf{week} \text{ vs } \$4{,}000/\text{week}$$

(Again, using fractions, we get 20,000 $/week instead of 20,160)
The net gain is now $16,160 per week, which is less than the previous two alternatives (from parts (b) and (c) above). All are better than the current situation.

2. Queueing Theory in Court (Based on an actual case) **

A worker in the State Unemployment Office is responsible for processing a company's forms when it opens for business. The worker can process an average of 4 forms per week. In 1992 an average of 1.8 companies per week submitted forms for processing, and the worker had an order flow time of 0.45 weeks (i.e. average time per form in the system). In 1993, an average of 3.9 companies per week submitted forms for processing and the worker had a 5 week order flow time. The poor worker was fired, but later sued to get her job back. The court ruled that because the amount of work submitted to the worker had approximately doubled, the worker's backlog (avg. time per form) also should have doubled. Because her backlog increased by more than a factor of 10, she must have been slacking off, so the state was justified in firing her. Assume exponential service times and Poisson arrivals, and a FCFS (first come, first served) queueing discipline.

Use the concepts learned in POM to defend the worker.

The judgement of the court was that the worker had been slacking off, meaning that her service rate (μ) had decreased with the increased workload (as λ increased from 1.8 to 3.9 forms/week). To defend the worker, we must compute the new service rate and show that it is at least 4 forms/week.

We know that $\lambda = 3.9$ forms/week and $W = 5$ weeks, and we want to solve for μ.

$$W = W_q + \tau = \tau \frac{\rho}{(1-\rho)} \frac{1}{2}(CV_A^2 + CV_S^2) + \tau = \tau \frac{\rho}{(1-\rho)} + \tau$$

Then, using $\rho = {}^{\lambda}/_{\mu}$ and $\tau = {}^{1}/_{\mu}$:

$$W = \frac{1}{\mu} . \frac{\lambda/\mu}{(1-\frac{\lambda}{\mu})} + \frac{1}{\mu} = \frac{1}{\mu} . \frac{\lambda/\mu}{(\mu-\lambda)/\mu} + \frac{1}{\mu} = \frac{\lambda}{\mu(\mu-\lambda)} + \frac{1}{\mu} = \frac{1}{\mu-\lambda}$$

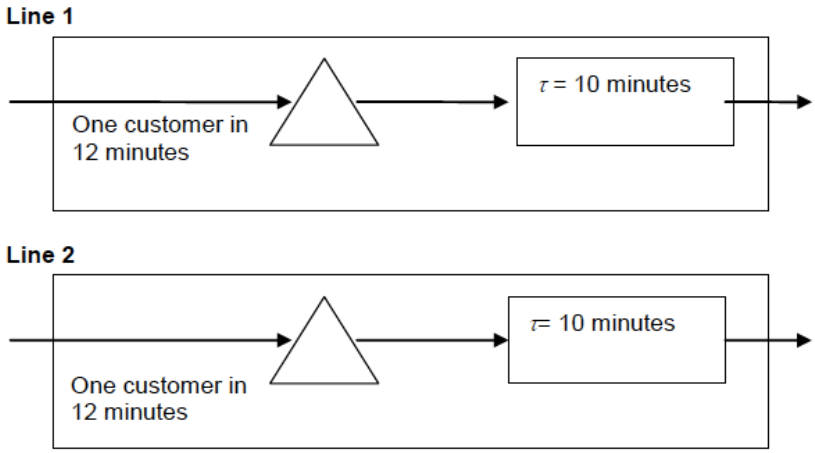
$$5 = \frac{1}{\mu - 3.9} \Rightarrow \mu = 4.1\ forms/week$$

Therefore, the service rate of the worker had actually increased, she should have not been fired, she should have been given a raise!

<u>Note</u>: We didn't need to use the information given for the year 1992, because we were already given 4 forms/week as the average service rate. Using that information does not change the answer: you find 4.02 forms/week as the worker's previous average service rate and it still increased to 4.1 forms/week.

3. Service Facility Design

A service organization has the following process design. There are two separate but identical service centers (with one employee each) that serve two distinct geographic areas. Each center has random service time with a mean of 10 minutes and a standard deviation of 10 minutes. Customers arrive at each center with an inter-arrival time of 12 minutes. The standard deviation of this inter-arrival time is 12 minutes.



a) *What is the current average wait time in minutes?**

Solution: $\tau = 10$ *minutes;* $\lambda = 1/12$; $s=1$

Utilization $\tau^*\lambda = \rho = 10/12=0.83$

$CV_A = 12/12 = 1$; $CV_s = 10/10 = 1$

$$W_q = \tau \frac{\rho}{(1-\rho)} \frac{1}{2}\left(CV_A^2 + CV_S^2\right) = 10 * .833/(1-.833) * (1^2+1^2)/2 = 50 \text{ minutes} \quad (W = 60 \text{ min.})$$

b) *The organization is considering strategies to reduce wait times. In particular, it is considering reducing service time from 10 minutes to 5 minutes. In this case, the standard deviation of service time would also be reduced from 10 minutes to 5 minutes. What is the average wait time in minutes if this strategy is implemented?**

$\tau = 5$; $\rho = 5/12 = .417$

$CV_s = 5/5 = 1$

$W_q = 5 * (.417)/(1-.417) * (1^2+1^2)/2 = 3.57$ min $\quad (W = 8.57$ min.)

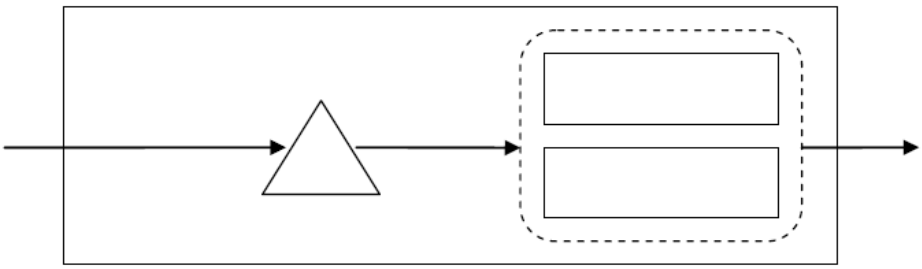
c) *Another strategy to reduce waiting time involves standardizing the activity, which would reduce the standard deviation of the service time from 10 to 2.5 minutes. Service time of 10 minutes will remain unchanged. What is the average wait time if this strategy is implemented?**

$\rho = 10/12 = .833$

$CV_s = 2.5/10 = .25$

$W_q = 10 * .833/(1-.833) * (1^2+.25^2)/2 = 26.49$ min $\quad (W = 36.49$ min)

d) The final strategy considered is to merge the two centers as shown below (in this case, the standard deviation of the inter-arrival times will be 6 minutes). What is the average wait time if this strategy is implemented?*



τ= 10 minutes, λ = 2/12=1/6; s=2

CV_a = 6/6 = 1; CV_s = 10/10 =1

*ρ = 10/ (6*2) = .833*

$$W_q = \frac{\tau}{s}\frac{1}{1-\rho}\frac{1}{2}\left(CV_a^2 + CV_s^2\right)\rho^{(\sqrt{2(s+1)}-1)}$$

$$W_q = 22.97\ min$$